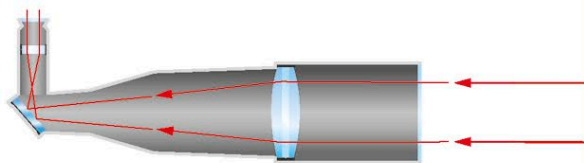
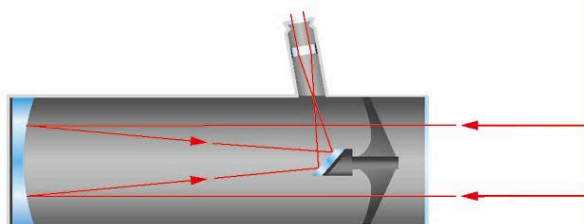


Types of telescope

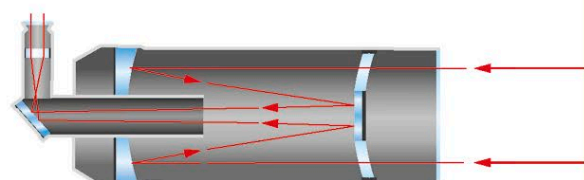
Refracting telescopes use lenses to gather and focus light. Reflecting telescopes do the same with mirrors—huge space telescopes use very large mirrors. Compound telescopes combine the best of lenses and mirrors.

**Refracting telescope**

A large convex lens focuses light rays to a mirror that reflects the light into the eyepiece, where a lens magnifies the image. Lenses refract the light, causing color distortion.

**Reflecting telescope**

A concave mirror reflects and focuses light to a secondary mirror, which reflects it into an eyepiece, where a lens magnifies the image. There is no color distortion.

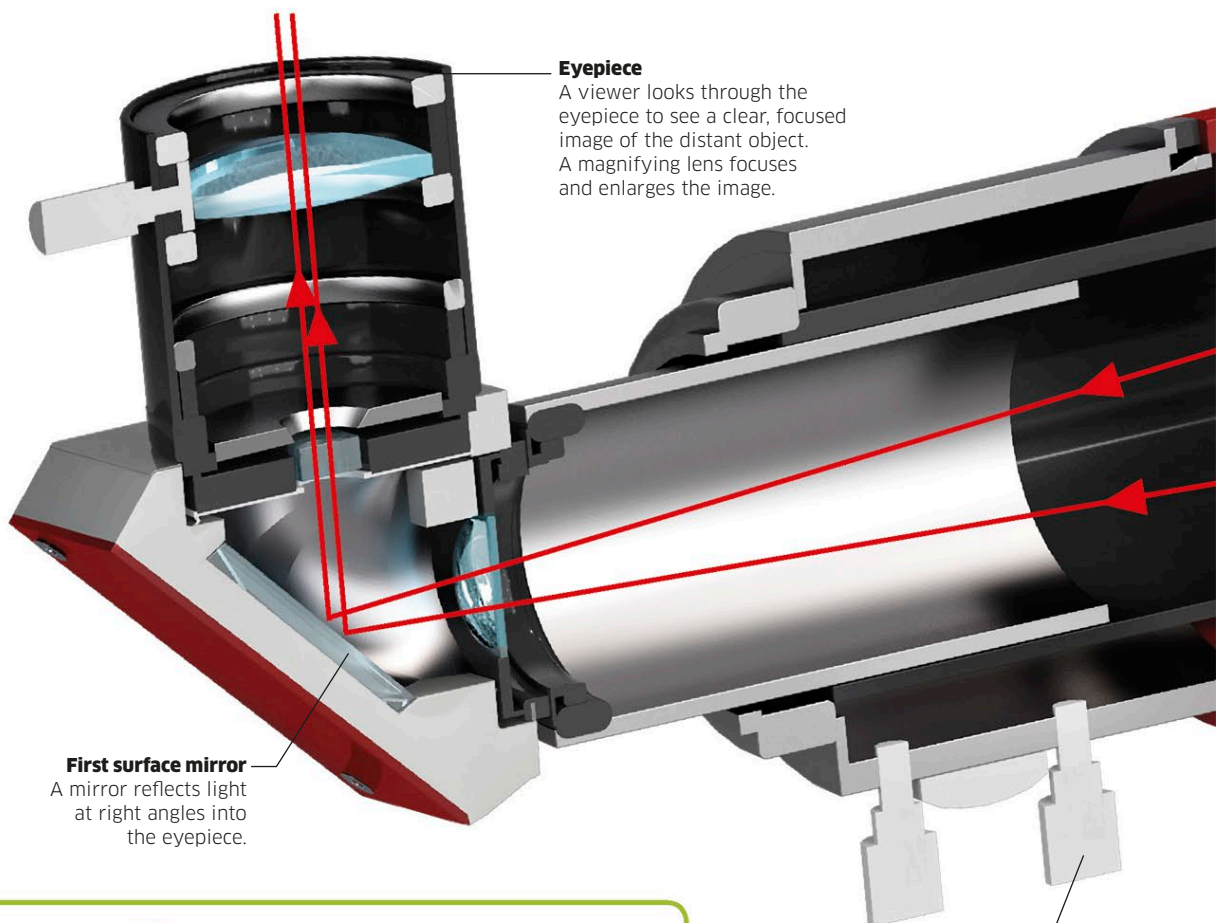
**Compound telescope**

The most common type of telescope, this combines lenses and mirrors to maximize magnification and eliminate distortion.

Telescopes

Powerful telescopes make faint objects, such as distant stars and galaxies, easier to see. They work by first gathering as much light as they can, using either a lens or a mirror, and then focusing that light into a clear image.

There are two main types of telescope: refracting, which focus light using lenses, and reflecting, which focus light using mirrors. Optical telescopes see visible light, but telescopes can also look for different kinds of electromagnetic radiation: radio telescopes receive radio waves and X-ray telescopes image X-ray sources. Telescopes use large lenses compared to microscopes, which are used to look at things incredibly close up, while binoculars work like two mini telescopes side by side.

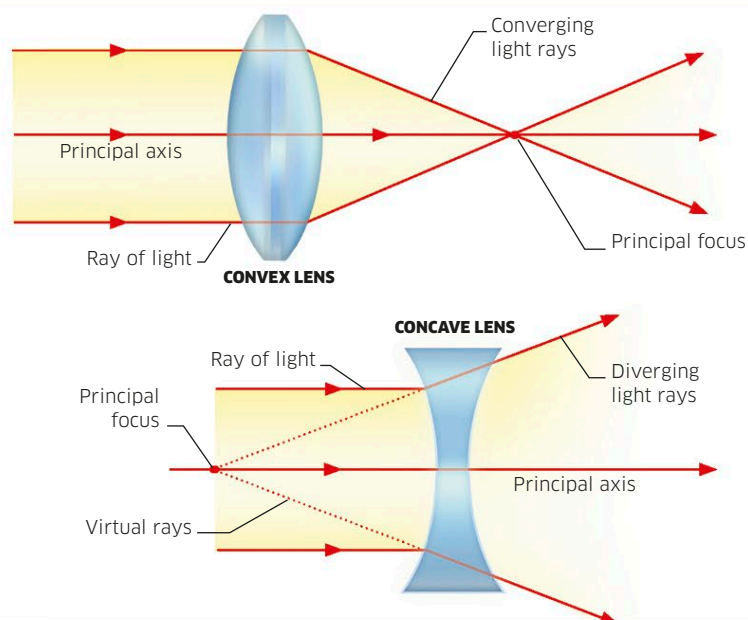
**Eyeiece**

A viewer looks through the eyeiece to see a clear, focused image of the distant object. A magnifying lens focuses and enlarges the image.

First surface mirror
A mirror reflects light at right angles into the eyepiece.

Convex and concave lenses

Convex, or converging, lenses take light and focus it into a point behind the lens, called the principal focus. This is the type of lens used in the glasses of a short-sighted person. By contrast, concave, or diverging, lenses spread light out. When parallel rays pass through a concave lens, they diverge as if they came from a focal point—the principal focus—in front of the lens.



Focus knob
Twisting the knob adjusts the focal length to focus the image.

A GERMAN-DUTCH LENS MAKER CALLED **HANS LIPPERSHEY** DEVELOPED THE EARLIEST REFRACTING TELESCOPE IN THE YEAR 1608. GALILEO IMPROVED THE DESIGN.

Isaac Newton
created the first reflecting telescopes to get around the problem of color distortion.